Musculoskeletal

Scoliosis



- Lateral spinal curve with rotation.
- Uneven shoulders/hips; rib prominence on forward bend.
- Radiographs measure Cobb angle.
- Observation, brace or surgery depends on curve and growth.
- Brace adherence and skin checks; support body image.

Torticollis



- Tight sternocleidomastoid → head tilt with rotation.
- Early physical therapy and gentle prescribed stretching.
- Encourage turning both ways; assess head shape.

Plagiocephaly



- Flattening from external pressure.
- Supervised awake tummy time; vary head position.
- Always place infant supine for sleep.
- Helmet in selected cases; assess for craniosynostosis.

Craniosynostosis



- Early fusion of skull suture(s).
- Abnormal head shape/growth.
- Craniofacial evaluation; surgery may be needed.

Clubfoot



- Foot turns inward/downward.
- Ponseti serial casts; often Achilles tenotomy, then brace.
- Check skin, toes and circulation; follow brace schedule.

Developmental Dysplasia of the Hip (DDH)



- Shallow/unstable hip joint.
- Limited abduction, unequal leg length; infant screening.
- Ultrasound or radiographs by age.
- Pavlik harness/cast or surgery as ordered.
- Do not adjust harness straps; skin/neurovascular checks.
- Use hip-safe swaddling, allowing flexion/abduction.





Slipped Capital Femoral Epiphysis



- Adolescent growth-plate slip; hip/groin or referred knee pain, limp.
- Limited internal rotation; leg may turn outward.
- Stop weight bearing; urgent orthopedic assessment.
- Surgical fixation prevents further slip.

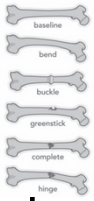
Legg-Calve-Perthes Disease



- Loss of blood supply to femoral head in childhood.
- Limp, hip/knee pain, restricted hip motion.
- Imaging; preserve motion and femoral-head containment.
- Activity/weight-bearing plan, therapy; selected surgery.

Neuromuscular Malformations

Fractures



- Pain, swelling, deformity or reduced use.
- Immobilize; assess distal color, warmth, pulse, sensation and movement.
- Growth-plate injuries can affect future growth.
- Open fracture: cover and seek urgent treatment.
- Reassess pain and neurovascular status after reduction/casting.

Casting



- Support wet plaster with palms; keep dry.
- Never insert objects or powders.
- Elevate for routine initial swelling as ordered.
- Report escalating pain, numbness, cool/pale digits or restricted movement.

Traction

- Maintain prescribed alignment and pull.
- Weights hang freely; ropes stay in pulleys.
- Skin/pin care and neurovascular checks; do not remove weights without an order.



Complications

- Compartment syndrome: severe unrelieved pain, pain with passive stretch, paresthesia.
- URGENT surgical assessment; pulse loss is late.
- Monitor skin injury, infection and immobility effects.

Muscular Dystrophy



- Progressive muscle weakness; Duchenne commonly affects boys.
- Gowers maneuver: uses hands to climb up legs when standing.
- Monitor respiratory and cardiac function.
- Preserve mobility; stretching, assistive devices, prescribed therapies.
- Support family, nutrition and genetic counseling.



Juvenile Idiopathic Arthritis







- Persistent inflammatory arthritis beginning in childhood.
- Pain, swelling, morning stiffness; some types have fever/rash.
- Balance rest with range-of-motion and low-impact activity.
- NSAIDs, disease-modifying drugs/biologics as prescribed.
- Regular eye exams for silent uveitis.
- Monitor growth, function and medicine effects.